

A Kinematic Model for Swarm Finite-Time Trajectory Tracking

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Abstract—This paper focuses on the trajectory tracking problem for a swarm of mobile agents. A kinematic model describing the interactions and evolutions of the swarm members is proposed and its main properties are analyzed emphasizing that the agents centroid is ensured to track in finite-time a given reference trajectory and that the agents reach an aggregation in finite-time in a hyper-ball moving around the centroid path. One of the main characteristics of the model is the presence of an interaction matrix, between agents coordinates, which allows to define some properties of the swarm allowing the creation of different forms of agents aggregations, i.e., spheres, ellipsoids, straight lines, etc. Indeed swarm properties related to the agents configuration around the performed path along with agents interactions and absence of collisions are analyzed depending on the chosen interaction matrix.

Index Terms—Finite-time control, multiagent system, swarm evolution, trajectory tracking.

I. INTRODUCTION

THE MULTIAGENT systems, like swarms and swarm robotics, are attracting the research attention especially in the last years. When entities, like animals or, in a general point of view, agents, are said to be a swarm then they exhibit a collective behavior consisting in aggregating together, for instance milling about the same spot or moving and migrating in some direction. Swarming is a very interdisciplinary topic [1], and swarming strategies apply to insects, birds, bacteria, and all the animals that exhibit collective behaviors. In particular, the term flocking is usually used to refer specifically to swarm behavior in birds [2], herding to refer to swarm behavior in tetrapods, and shoaling or schooling to refer to swarm behavior in fish [3]. Nowadays the term swarm is applied also to inanimate entities which exhibit parallel behaviors, like in a set of robots, an earthquake swarm, or a swarm of stars.

Swarm behavioral analysis is usually performed from an abstract point of view or focusing on specific applicative contexts. Studies on swarm agents interactions with no context

particularizations have been performed by [4]–[7] where a deep analysis on swarm formation properties under interactions model assumptions is proposed. In particular, the authors assume a swarm model where each agent has complete knowledge about all other agents locations, and steady state properties and aggregations capabilities of the swarm are proved. Further behavioral analysis can be found in [8]–[10] where agents agreement in presence of antagonistic interactions, modeled as negative weights on the communication graph, is studied showing that also under this communication topology, all agents can converge to a consensus value which is the same for all agents except for the sign.

A recent survey on distributed cooperative control problems of multiagent systems on the basis of a variety of sampling mechanisms can be found in [11]. Other recent results about multiagent system with time delay, uncertainties, and linear feedback are provided in [12].

Starting from this abstract point of view, swarm evolution inspires significant interests in various specific research fields, showing its adaptability in clustering [13], vehicles or robots moving on the ground [14]–[20], and underwater or flying in the sky [21], [22]. In a swarm of agents, all the entities are able, in an unconscious way, to aggregate while an emergent global behavior arises [23]. Examples of collective behaviors are formation shape control or agents trajectory tracking and also consensus reaching over network of interconnected agents. In [24], a novel concept of formation learning control is introduced for the formation control of a group of underwater vehicles. A distributed formation learning control scheme formed by an adaptive observer and a decentralized deterministic learning controller is described. Using the proposed control scheme the formation learning control protocol interests each local agent in a distributed way where no global information is required.

In particular, in [25] and [26] the consensus leader–follower problem for a multiagents system is studied. The first work analyzes a cooperative consensus tracking in the presence of a single high-dimensional leader the dynamics of which are allowed to be different with respect to the ones of the followers. In the second work, the authors face the leader-following consensus for a class of second-order multiagent systems with nonlinear dynamics. In the consensus context, the collective behavior has been studied also in [27] where a two-stage optimization method including primal decomposition and distributed consensus is proposed for a class of distributed convex optimization problems providing a stability analysis for the proposed algorithm.

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Among the many other works that study the collective behavior in a leader-follower scheme, in [28] the tracking control problem for a group of multiple mobile robots moving along straight-line paths is studied. The authors assume that the information about the reference trajectory is available only to one swarm leader robot which sends it to other intermediary leaders, forming a hierarchical communication scheme like a simple spanning directed tree, that is, the minimal configuration possible to achieve the formation-tracking objective. Partially linear time-varying controllers with an added nonlinearity are designed separating the tasks of position-tracking and orientation, resulting in uniform global asymptotic stabilization of the closed-loop robotic system. In [29], swarm tracking problems with group dispersion and cohesion behaviors are discussed; the swarm is divided into a leader group, the members of which have knowledge about the desired generalized coordinates, and a follower group guided by the leader group. Distributed tracking of generalized coordinate derivatives is ensured in both cases of constant and time-varying generalized coordinate derivatives.

In this paper, a kinematic model for a swarm evolution is proposed. The swarm members are all driven by a common signal ruling the swarm centroid reference path. The interactions among agents follow a simple model and they can be mainly tuned by a time invariant interaction matrix. Depending on this matrix, the model can balance attraction and repulsion among agents so that to ensure swarm aggregation and collision avoidance among swarm members, respectively. It will be shown that the swarm centroid will reach the imposed reference trajectory in finite-time and the swarm will converge into a bounded region around the swarm centroid in finite-time too. Therefore, finite-time concepts address the lack of velocity of the swarm cohesion and the centroid reference tracking.

It will be also shown that the amplitude of this region can be chosen *a-priori* by acting on the interaction matrix and these dependencies will be studied highlighting the simplicity in designing the interaction matrix and the swarm evolution to accomplish a given task. Moreover, the steady state swarm configuration will be analyzed and a set of useful properties will be given. Agents collisions study will be performed too relating the clashes avoidance with the interaction matrix properties. Finally, considerations about the proposed kinematic model in the cases of finite-size agents or limited connectivity among swarm elements will be provided.

The proposed agents evolution has to be intended as a higher-level model ignoring lower-level dynamics and it is relevant because it allows for studying swarms behavior in an independent way from detailed agents. Moreover the resulting path, for each modeled agent, can be used as a reference trajectory for the actual agents tracking control law [30]. Other results on finite-time control but related to more specific dynamic agents models have been presented in [31] and [32] where a finite-time cooperative controller for the consensus problem in a team of multiple nonholonomic mobile robots has been designed and a finite-time synchronization strategy for a class of second-order nonlinear multiagent

systems with a leader-follower architecture has been proposed.

Preliminary results about this line of research have been presented in [33] where a less general swarm model has been analyzed showing its main properties.

The remainder of this paper is organized as follows. Section II describes the proposed kinematic model for the swarm evolution; in Sections III and IV, the swarm finite-time trajectory tracking properties and the agents collision avoidance are analyzed, respectively; in Section V, the cases of finite-size agents and limited connectivity among agents are discussed; and in Section VI, numerical results are proposed and Section VII is devoted to the conclusions.

Notations: The following conventions and notations are used throughout this paper.

- 1) $\mathbb{R}$ is the set of real numbers.
- 2) $\mathbb{R}^k$ denotes the set of k -dimensional vectors of real numbers.
- 3) $\mathbb{R}^{k \times k}$ denotes the set of $k \times k$ -dimensional matrices of real numbers.
- 4) $\mathbb{N}^+$ is the set of integer numbers greater than 0.
- 5) $\mathcal{C}^1$ is the class of all differentiable functions whose first derivative is continuous.

II. SWARM INTERACTION MODEL

In this section, the proposed swarm interaction model is described. It represents an extension to the model proposed in [33] with a modified attraction/repulsion term introducing new degrees of freedom in the swarm formation.

Let us consider a swarm composed by n agents in a k -dimensional Euclidean space with a fixed network topology described by an undirected fully connected graph. This means that all agents know the exact position of all other members. Moreover, they are considered as ideal points moving simultaneously without delay [4]–[6]. Some considerations on how to relax these assumptions will be provided later.

The motion dynamics evolve in continuous time. The position of the i th agent, at time t , is denoted by $z_i(t) \in \mathbb{R}^k$. The interaction dynamic for agent i is given by

$$\dot{z}_i(t) = -z_i(t) - \frac{1}{n} \sum_{j=1}^n M \frac{z_i(t) - z_j(t)}{\|z_i(t) - z_j(t)\|^v + \varepsilon} + r(t) + \dot{r}(t) - \zeta(t) \quad (1)$$

where $v \in \mathbb{N}^+$, $\varepsilon > 0$, and:

- 1) $M \in \mathbb{R}^{k \times k}$ is a full-rank constant matrix which rules the interactions between agents coordinates;
- 2) $c(t) = (1/n) \sum_{j=1}^n z_j(t)$ is the swarm centroid;
- 3) $r(t) \in \mathbb{R}^k$, assumed to be $r(t) \in \mathcal{C}^1$, is the reference trajectory for the swarm centroid $c(t)$ and it is the same for all the agents;
- 4) $\zeta(t) \in \mathbb{R}^k$ and its s th component is computed from the s th components of $c(t)$ and $r(t)$ as

$$\zeta_s(t) = \text{sign}(c_s(t) - r_s(t)) |c_s(t) - r_s(t)|^\beta, \quad \forall s = 1, \dots, k$$

with $\beta \in (0, 1)$.

The interaction among agents in the kinematic model (1), i.e., the nonlinear term $[x/(\|x\|^v + \epsilon)]$ (with $v = 1$) is used in other applicative contexts too (see [34], [35] where the problem of designing a distributed algorithm for multiagent systems to track the average of multiple reference signals, is addressed).

A. Model Decentralization

Model (1) is formulated considering that each agent knows its own position with respect to the same reference frame of $r(t)$. This hypothesis is acceptable, for example, in a simple swarming model for real multirobot system in an environment where the GPS is a reliable positioning system [36], [37]. This means a scenario where a swarm of robots is moving within an outdoor environment according to the commands provided by a centralized station. In all cases where it is necessary to decentralize the information and then to run the swarming algorithm on the process unit of each agent, a decentralized model is mandatory.

The proposed model can be decentralized considering a real external agent evolving as imposed by $r(t)$. This agent does not affect the dynamics of the swarm. For this reason this new agent has to be maneuvered in order to evade capture or being enclosed by the swarm of agents so that the overall swarm evolution results as in model (1) [30]. In this context, each agent evolves according to the described dynamics computing all the information in its local reference frame.

Let us define for each agent i a variable $\Delta_i(t) \triangleq z_i(t) - r(t)$ representing the distance of the agent from the reference trajectory at time t , the model (1) becomes

$$\dot{\Delta}_i(t) = -\Delta_i(t) - \frac{M}{n} \sum_{j=1}^n \frac{\Delta_i(t) - \Delta_j(t)}{\|\Delta_i(t) - \Delta_j(t)\|^v + \epsilon} - \zeta(t) \quad (2)$$

where the s th component of $\zeta(t)$ is

$$\zeta_s(t) = \text{sign}(\sigma_s(t)) |\sigma_s(t)|^\beta, \forall s = 1, \dots, k$$

and $\sigma_s(t)$ is the s th component of the signal

$$\sigma(t) = c(t) - r(t) = \frac{1}{n} \sum_{i=1}^n \Delta_i(t). \quad (3)$$

The resulting overall formulation (2) uses only local and decentralized information related to the displacement between each agent and the agent evolving as imposed by the reference trajectory $r(t)$. In a real scenario, each swarm element can use the signal $\Delta_i(t)$ as a reference trajectory for its displacement from $r(t)$. In other words, as shown in Fig. 1, a control scheme can be designed for each agent where its measured displacement from $r(t)$, available by standard local sensors placed on the agent, is fed back to a properly chosen control law and forced to follow $\Delta_i(t)$.

B. Agents Interactions

The interaction function representing the function of attraction/repulsion between agents is modeled by the term

$$-\frac{1}{n} \sum_{j=1}^n M \frac{z_i(t) - z_j(t)}{\|z_i(t) - z_j(t)\|^v + \epsilon}.$$

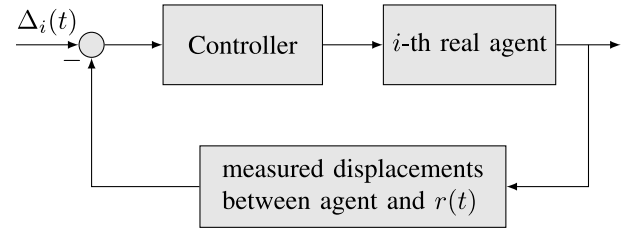


Fig. 1. Displacement control scheme for the i th real agent.

To clarify the behavior of this term, consider a swarm formed by a couple of agents i and j moving along one direction (i.e., x) evolving as

$$\begin{aligned} \dot{x}_i(t) &= -\frac{m}{2} \frac{x_i(t) - x_j(t)}{|x_i(t) - x_j(t)|^v + \epsilon} \\ \dot{x}_j(t) &= -\frac{m}{2} \frac{x_j(t) - x_i(t)}{|x_j(t) - x_i(t)|^v + \epsilon} \end{aligned}$$

where $m \in \mathbb{R}$. The difference $\xi(t) = x_i(t) - x_j(t)$ between agents locations is then expressed as

$$\dot{\xi}(t) = -m \frac{\xi(t)}{|\xi(t)|^v + \epsilon}.$$

This term can be attractive or repulsive depending on the parameter m . In order to justify this fact, let us consider the Lyapunov function $V_{i,j}(t) = (1/2)\xi^2(t)$. Its time derivative is

$$\dot{V}_{i,j}(t) = \xi(t)\dot{\xi}(t) = -m \frac{\xi^2(t)}{|\xi(t)|^v + \epsilon}$$

from which it follows that the sign of $\dot{V}_{i,j}(t)$ is $-\text{sign}(m)$; therefore, depending on m , the term $\xi(t)$ decreases to zero or it diverges, i.e., the interaction behavior is attractive or repulsive, respectively. Moreover, the parameters v and ϵ modulate the interaction factor. Fig. 2 shows four cases of the interaction function by choosing $m = -1$, $v = 1, 2$, and $\epsilon = 0.01, 0.0001$. If $v = 1$ then the interaction effect tends to the $\text{sign}(\cdot)$ function as $\epsilon \rightarrow 0^+$. On the contrary, if $v > 1$, as v grows the interaction tends to zero for $|\xi(t)| \rightarrow \infty$ and when $\epsilon \rightarrow 0^+$, the interaction term for $|\xi(t)| \rightarrow 0^+$ is increased, approximating an *unbounded-like* behavior.

The interaction imposed by the matrix M can be easily tuned to modulate the attraction/repulsion among agents on the k axes. In particular if $M = \mu I_k$, $\mu \in \mathbb{R}$, where I_k is the identity matrix of order k , the interaction is uniformly distributed on the k dimensions. Otherwise, if M is chosen to represent a repulsive effect, it is straightforward to show that the greater is the module of one of its eigenvalues the higher will be the repulsion on the related eigenspace.

To the best of authors' knowledge, the most similar model to the proposed one has been presented in [4]. Model (1) mainly differentiates in the agents interaction term which can be attractive or repulsive and it is not formulated as the sum of the two separate effects. The interactions effects can be also tuned by properly choosing the matrix M which gives some degrees of freedom on the final agents configuration. Moreover, the presented kinematic evolution ensures finite-time swarm centroid reference tracking as explained in the next section.

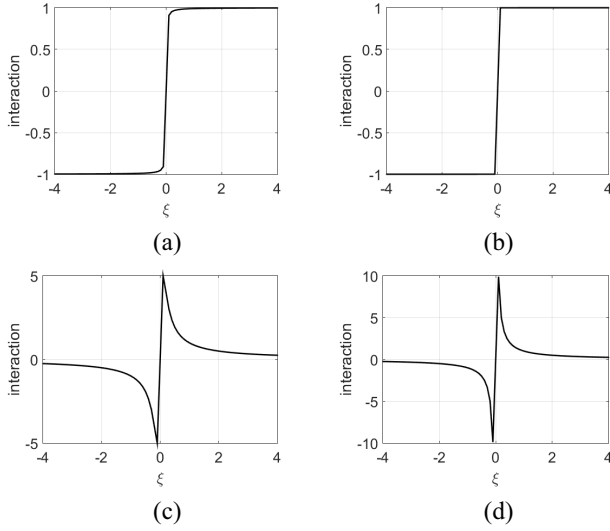


Fig. 2. Interaction effect with $\nu = 1, 2$ and $\varepsilon = 0.01, 0.0001$.

III. TRAJECTORY TRACKING

In this section, the swarm behavior is analyzed looking at the swarm centroid evolution (swarm macroscopic level) and studying the agents aggregation properties (swarm microscopic level).

Theorem 1: The swarm centroid $c(t)$ converges to the trajectory $r(t)$ in finite-time.

Proof: Let $u(t) \triangleq r(t) + \dot{r}(t) - \zeta(t)$, then

$$\begin{aligned} \dot{\sigma}(t) &= \frac{1}{n} \sum_{i=1}^n \dot{z}_i(t) - \dot{r}(t) \\ &= \frac{1}{n} \sum_{i=1}^n \left(-z_i(t) - \frac{1}{n} \sum_{j=1}^n \frac{M(z_i(t) - z_j(t))}{\|z_i(t) - z_j(t)\|^\nu + \varepsilon} + u(t) \right) \\ &\quad - \dot{r}(t) = -\frac{1}{n} \sum_{i=1}^n z_i(t) \\ &\quad - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \frac{M(z_i(t) - z_j(t))}{\|z_i(t) - z_j(t)\|^\nu + \varepsilon} + u(t) - \dot{r}(t) \end{aligned}$$

where $\sigma(t)$ is defined in (3).

Considering that the attraction/repulsion force between any pair of agents i and j is anti-symmetric in i and j

$$\sum_{i=1}^n \sum_{j=1}^n M \frac{z_i(t) - z_j(t)}{\|z_i(t) - z_j(t)\|^\nu + \varepsilon} = 0_k$$

where 0_k is the origin of $\mathbb{R}^k$, and then

$$\dot{\sigma}(t) = -\sigma(t) - \zeta(t). \quad (4)$$

Equation (4) represents a set of k independent equations, one for each component of $\sigma(t)$

$$\dot{\sigma}_j(t) = -\sigma_j(t) - \text{sign}(\sigma_j(t)) |\sigma_j(t)|^\beta, \quad j = 1, \dots, k.$$

To proof the convergence to zero of each component $\sigma_j(t)$, let us consider the Lyapunov function $\Xi_j(t) = (1/2)\sigma_j^2(t)$, the

derivative of which can be expressed as

$$\begin{aligned} \dot{\Xi}_j(t) &= \sigma_j(t) \dot{\sigma}_j(t) = \sigma_j(t) \left(-\sigma_j(t) - \text{sign}(\sigma_j(t)) |\sigma_j(t)|^\beta \right) \\ &= -\sigma_j^2(t) - |\sigma_j(t)|^{\beta+1} \leq -|\sigma_j(t)|^{\beta+1} \\ &= -2^{\frac{\beta+1}{2}} \frac{|\sigma_j(t)|^{\beta+1}}{2^{\frac{\beta+1}{2}}} = -2^{\frac{\beta+1}{2}} \left(\frac{\sigma_j^2(t)}{2} \right)^{\frac{\beta+1}{2}}. \end{aligned}$$

By expliciting the dependence of $\dot{\Xi}_j(t)$ from $\Xi_j(t)$, it follows that:

$$\dot{\Xi}_j(t) \leq -2^{\frac{\beta+1}{2}} \Xi_j(t)^{\frac{\beta+1}{2}} = -\alpha \Xi_j^\eta(t)$$

where $\eta = [(\beta + 1)/2]$ and $\alpha = 2^\eta$. Finally,

$$\begin{cases} \Xi_j^{1-\eta}(t) \leq \Xi_j^{1-\eta}(t_0) - \alpha(1-\eta)(t-t_0), & t \in [t_0, \tau_j] \\ \Xi_j(t) = 0, & \forall t \geq \tau_j \end{cases}$$

where t_0 is the initial time and $\tau_j = t_0 + \frac{\Xi_j^{1-\eta}(t_0)}{\alpha(1-\eta)}$.

Therefore, each component $\sigma_j(t)$ goes to zero in finite-time stating that each component of the swarm centroid goes to the related component of $r(t)$ in finite-time too after at most τ_j . ■

Remark 1: Note that the solution to finite-time tracking problem for multiagent systems addressed in this paper does not guarantee that the settling time is independent on the initial states of the agents. This aspect should be considered in the practical applications where the appointed finite-time has to be offline preassigned according to task requirements (see [38] where the problem of appointed-time for consensus is discussed).

A. Analysis of Swarm Cohesion

In the next theorem, it will be proved that the swarm converges toward a bounded region around the centroid trajectory whose size is a function of the matrix M and of the parameters ν and ε . Moreover the convergence takes place in finite-time. This means that the agents aggregate in finite-time in a hyperball whose center, i.e., the swarm centroid, tracks the reference trajectory in finite-time too. Note that the proof follows the same argument as in [33] (Theorem 2), where the Lyapunov candidate function now takes into account the presence of the interaction matrix M and of the parameter ν . The new region toward which the swarm is converging depends on the matrix M , thus providing some degrees of freedom in terms of the shape of the aggregation region.

Lemma 1: Let f be a function such that $f: \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$, then

$$h(f) = \frac{f}{f^\nu + \varepsilon} \leq \frac{1}{\nu} \left(\frac{\nu-1}{\varepsilon} \right)^{\frac{\nu-1}{\nu}}, \quad \varepsilon > 0, \quad \nu \in \mathbb{N}^+. \quad (5)$$

Proof: Note that for $\nu = 1$, one has $[f/(f + \varepsilon)] < 1$. For $\nu > 1$, $h(f)$ has the stationary point in $\tilde{f} = ([\nu - 1]/\varepsilon)^{-(1/\nu)}$. Moreover $\lim_{f \rightarrow \tilde{f}} (d^2 h / df^2) = -(\varepsilon/\nu)(\varepsilon/[\nu - 1])^{-2-(1/\nu)}$. Therefore, (5) follows. ■

Theorem 2: Consider the swarm model (1) with a generic matrix $M \in \mathbb{R}^{k \times k}$, then all the agents of the swarm move

toward and remain within a hyper-ball centered on the reference trajectory

$$\mathcal{B} = \{z \in \mathbb{R}^k : \|z - r(t)\| \leq \rho\}$$

where $\rho = (\|M\|/\nu)([v-1]/\varepsilon)^{([v-1]/\nu)}$.

Furthermore, all agents will move into the hyper-ball $\mathcal{B}$ in finite-time specified by

$$t = \max_{i=1,\dots,n} \bar{t}_i$$

where

$$\bar{t}_i = t_0 - \frac{n}{2} \log \left(\frac{\rho^2}{\|z_i(t_0) - r(t_0)\|^2} \right)$$

and t_0 is the initial evolution time.

Proof: The centroid swarm $c(t)$ reaches in finite-time, namely $\tau_c = \max_{j=1,\dots,n} \tau_j$, the reference trajectory $r(t)$ (as stated in the previous theorem). This means that for $t \geq \tau_c$ the agents evolution is

$$\dot{z}_i(t) = -z_i(t) - \frac{1}{n} \sum_{j=1}^n M \frac{z_i(t) - z_j(t)}{\|z_i(t) - z_j(t)\|^v + \varepsilon} + r(t) + \dot{r}(t) \quad (6)$$

since $\zeta(t) = 0$ for $t \geq \tau_c$.

Define the distance between each agent i and the trajectory $r(t)$ as $g_i(t)$, i.e., $g_i(t) \triangleq z_i(t) - r(t)$, then (6) becomes

$$\dot{g}_i(t) = -g_i(t) - \frac{1}{n} \sum_{j=1}^n M \frac{g_i(t) - g_j(t)}{\|g_i(t) - g_j(t)\|^v + \varepsilon}. \quad (7)$$

Using (7), it can be shown that agents asymptotically enter into the hyper-ball

$$\tilde{\mathcal{B}} = \{z \in \mathbb{R}^k : \|z\| \leq \gamma\}$$

where $\gamma = \|M\|[(n-1)/nv]([v-1]/\varepsilon)^{([v-1]/\nu)}$, by simply considering the Lyapunov function $G_i(t) = (1/2)g_i(t)^T g_i(t)$. The derivative of $G_i(t)$ is

$$\begin{aligned} \dot{G}_i(t) &= g_i^T(t) \dot{g}_i(t) \\ &= g_i^T(t) \left(-g_i(t) - \frac{1}{n} \sum_{j=1}^n M \frac{g_i(t) - g_j(t)}{\|g_i(t) - g_j(t)\|^v + \varepsilon} \right) \\ &= -\|g_i(t)\|^2 + \frac{1}{n} \sum_{j=1}^n \frac{g_i^T(t) M (g_j(t) - g_i(t))}{\|g_i(t) - g_j(t)\|^v + \varepsilon} \\ &\leq -\|g_i(t)\|^2 + \frac{1}{n} \|g_i(t)\| \|M\| \sum_{j=1}^n \frac{\|g_j(t) - g_i(t)\|}{\|g_i(t) - g_j(t)\|^v + \varepsilon}. \end{aligned}$$

Since (see Lemma 1)

$$\frac{\|g_j(t) - g_i(t)\|}{\|g_i(t) - g_j(t)\|^v + \varepsilon} \leq \frac{1}{v} \left(\frac{v-1}{\varepsilon} \right)^{\left(\frac{v-1}{v} \right)}$$

it follows that:

$$\begin{aligned} \dot{G}_i(t) &\leq -\|g_i(t)\|^2 + \frac{1}{n} \|g_i(t)\| \|M\| \sum_{j \neq i} \frac{1}{v} \left(\frac{v-1}{\varepsilon} \right)^{\left(\frac{v-1}{v} \right)} \\ &= -\|g_i(t)\|^2 + \|g_i(t)\| \|M\| \frac{n-1}{nv} \left(\frac{v-1}{\varepsilon} \right)^{\left(\frac{v-1}{v} \right)} \end{aligned}$$

and then

$$\dot{G}_i(t) < 0 \text{ if } \|g_i(t)\| > \|M\| \frac{n-1}{nv} \left(\frac{v-1}{\varepsilon} \right)^{\left(\frac{v-1}{v} \right)}$$

which states that each agent g_i enters into the hyper-ball $\tilde{\mathcal{B}}$.

Clearly, the swarm will also enter into a generic hyper-ball with radius greater than γ and, in particular, in that of radius ρ

$$\hat{\mathcal{B}} = \{z \in \mathbb{R}^k : \|z\| \leq \rho\}.$$

The rest of the proof is devoted to find the time needed for the agent to enter into the hyper-ball with radius ρ . To this aim, consider the previous defined Lyapunov function, its derivative can be also bounded as

$$\begin{aligned} \dot{G}_i(t) &\leq -\|g_i(t)\|^2 + \|g_i(t)\| \|M\| \frac{n-1}{nv} \left(\frac{v-1}{\varepsilon} \right)^{\left(\frac{v-1}{v} \right)} \\ &= -\|g_i(t)\|^2 + \frac{\|g_i(t)\|^2}{\|g_i(t)\|} \|M\| \frac{n-1}{nv} \left(\frac{v-1}{\varepsilon} \right)^{\left(\frac{v-1}{v} \right)}. \end{aligned}$$

If $g_i(t)$ is out of $\hat{\mathcal{B}}$ then

$$\|g_i(t)\| \geq \frac{\|M\|}{v} \left(\frac{v-1}{\varepsilon} \right)^{\left(\frac{v-1}{v} \right)}$$

which states that

$$\frac{1}{\|g_i(t)\|} \leq \frac{v}{\|M\|} \left(\frac{v-1}{\varepsilon} \right)^{-\left(\frac{v-1}{v} \right)}$$

and yields to

$$\begin{aligned} \dot{G}_i(t) &\leq -\|g_i(t)\|^2 \\ &\quad + \frac{v\|g_i(t)\|^2}{\|M\|} \left(\frac{v-1}{\varepsilon} \right)^{-\left(\frac{v-1}{v} \right)} \|M\| \frac{n-1}{nv} \left(\frac{v-1}{\varepsilon} \right)^{\left(\frac{v-1}{v} \right)} \\ &= -\|g_i(t)\|^2 + \frac{n-1}{n} \|g_i(t)\|^2 = -\frac{\|g_i(t)\|^2}{n}. \end{aligned}$$

This permits to bound $\dot{G}(t)$ in terms of $G(t)$, as

$$\dot{G}_i(t) \leq -\frac{2}{n} G_i(t)$$

which yields to

$$G_i(t) \leq G_i(t_0) e^{-\frac{2}{n}(t-t_0)}.$$

Therefore, the agent g_i enters into $\hat{\mathcal{B}}$ when

$$\rho^2 \geq 2G_i(t_0) e^{-\frac{2}{n}(t-t_0)} \rightarrow t \geq t_0 - \frac{n}{2} \log \left(\frac{\rho^2}{2G_i(t_0)} \right)$$

and, consequently, $z_i(t)$ enters into the hyper-ball $\mathcal{B}$ in finite-time. ■

Remark 2: Note that the bound ρ does not depend on the swarm size n . This means that if n increases the density of the

swarm increases too. Moreover, if $v = 2$ then the density is uniformly in the hyper-ball and although the distribution inside the swarm can vary and have fluctuations, uniform density of a swarm is often a good first-order approximation for many swarms [39].

B. Steady State Analysis

When agents enter in the hyper-ball $\mathcal{B}$, as discussed above, they may reach constant positions or exhibit continuous movement with respect to the centroid path. The latter is typical of some engineering applications such as uninhabited air vehicles where it may happen that agents never stop. It is also the case of vortexlike swarming behavior observed, for example, in a wide range of biological systems from bacteria to higher vertebrates [40]. In other contexts, however, it is necessary to guarantee fixed steady state configuration and/or global shape formation from geometric primitives based on the attractive and repulsive forces between the agents. Next theorem presents sufficient conditions for agents steady configuration as time goes to infinity.

Lemma 2 (LaSalle's Invariance Principle [41]): Consider the system $\dot{z} = \phi(z)$, where $\phi : D \rightarrow \mathbb{R}^l$ is a locally Lipschitz map from a domain $D \subset \mathbb{R}^l$ into $\mathbb{R}^l$. Let $\Omega \subset D$ be a compact set that is positively invariant with respect to the system dynamic. Let $V : D \rightarrow \mathbb{R}$ be a continuously differentiable function such that $\dot{V}(z) \leq 0$ in Ω . Let E be the set of all points in Ω , where $\dot{V}(z) = 0$. Let H be the largest invariant set in E . Then every solution starting in Ω approaches H as $t \rightarrow \infty$.

Theorem 3: If $M = M^T < 0$ and there exists a function U_i such that

$$\frac{\partial U_i}{\partial g_i} = M^{-1} \dot{g}_i(t) \text{ and } \frac{\partial U_i}{\partial g_j} = -\frac{1}{n} \frac{g_j(t) - g_i(t)}{\|g_i(t) - g_j(t)\|^v + \varepsilon}$$

where $g_i(t) \triangleq z_i(t) - r(t)$, then the agents will reach a fixed steady state position with respect to the reference trajectory $r(t)$, i.e., $\lim_{t \rightarrow \infty} \dot{g}_i(t) = 0_k$, $\forall i = 1, \dots, n$.

Proof: Let us consider (7) which models the agents when the swarm centroid is on the reference trajectory $r(t)$ (note that this condition occurs in finite-time).

If M is not singular each agent evolution can be written as

$$M^{-1} \dot{g}_i(t) = -M^{-1} g_i(t) - \frac{1}{n} \sum_{j=1}^n \frac{g_i(t) - g_j(t)}{\|g_i(t) - g_j(t)\|^v + \varepsilon}.$$

Consider the following scalar function:

$$U = \sum_{i=1}^n U_i - \frac{1}{2} \sum_{i=1}^n g_i(t)^T M^{-1} g_i(t)$$

then its derivative is

$$\begin{aligned} \dot{U} &= \sum_{i=1}^n \dot{U}_i - \sum_{i=1}^n g_i(t)^T M^{-1} \dot{g}_i(t) \\ &= \sum_{i=1}^n \sum_{j=1}^n \left(\frac{\partial U_i}{\partial g_j} \right)^T \dot{g}_j(t) - \sum_{i=1}^n g_i^T M^{-1} \dot{g}_i(t) \end{aligned}$$

which can be rewritten as follows:

$$\begin{aligned} \dot{U} &= \sum_{i=1}^n \dot{g}_i^T(t) M^{-1} \dot{g}_i(t) \\ &\quad - \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^n \frac{(g_j(t) - g_i(t))^T}{\|g_i(t) - g_j(t)\|^v + \varepsilon} \dot{g}_j(t) \\ &\quad - \sum_{i=1}^n g_i^T(t) M^{-1} \dot{g}_i(t). \end{aligned}$$

Noting now that

$$\begin{aligned} &-\frac{1}{n} \sum_{i=1}^n \sum_{j=1}^n \frac{(g_j(t) - g_i(t))^T \dot{g}_j(t)}{\|g_i(t) - g_j(t)\|^v + \varepsilon} \\ &= \sum_{j=1}^n \left(-\frac{1}{n} \sum_{i=1}^n \frac{g_j(t) - g_i(t)}{\|g_i(t) - g_j(t)\|^v + \varepsilon} \right)^T \dot{g}_j(t) \\ &= \sum_{j=1}^n \left(M^{-1} (\dot{g}_j(t) + g_j(t)) \right)^T \dot{g}_j(t) \\ &= \sum_{j=1}^n \dot{g}_j^T(t) M^{-1} \dot{g}_j(t) + \sum_{j=1}^n g_j^T(t) M^{-1} \dot{g}_j(t) \end{aligned}$$

the function $\dot{U}$ assumes the expression

$$\dot{U} = 2 \sum_{i=1}^n \dot{g}_i^T(t) M^{-1} \dot{g}_i(t) \leq 0.$$

According to Theorem 2, the agents are in the compact set $\mathcal{B}$ which is positively invariant with respect to the swarm dynamic. Then we can apply LaSalle's invariance principle (Lemma 2) to conclude that the swarm reaches a fixed steady state configuration according to the invariant set defined as $\{g_i(t) \mid \dot{U} = 0\} = \{g_i(t) \mid \dot{g}_i(t) = 0\}$. ■

Note that, unlike the case described in [42], where a team performance function formed by a sum of convex local objective functions is optimized and each agent utilizes only local interaction and the gradient of its own local objective function, the previous theorem states that, using the proposed kinematic model, the overall swarm behavior tends to minimize the function U , but the i th component U_i not only affects the evolution of the i th agent but also of the entire swarm.

Results stated in the previous theorem can be particularized for the cases $v = 1, 2$ by using

$$\begin{aligned} U_i &= -\frac{1}{2} g_i^T(t) M^{-1} g_i(t) \\ &\quad - \frac{1}{n} \sum_{j=1}^n (\|g_i(t) - g_j(t)\| - \varepsilon \log(\varepsilon + \|g_i(t) - g_j(t)\|)) \end{aligned}$$

for $v = 1$, and

$$U_i = -\frac{1}{2} g_i^T(t) M^{-1} g_i(t) - \frac{1}{2n} \sum_{j=1}^n \log(\varepsilon + \|g_i(t) - g_j(t)\|^2)$$

for $v = 2$.

IV. AGENTS COLLISION AVOIDANCE

In an agents formation control strategy, the collision avoidance is one of the desirable features. From a theoretical point of view the collision is avoided only if the interaction effect among agents is unbounded for infinitesimally small distances. This is not the case, unless $\epsilon \rightarrow 0$ and $\nu > 1$. We address the problem of collision avoidance by making some assumptions on the interactions among agents. The result is a simple condition on the parameters M , ν , and ϵ of the model (1).

Let $\mathcal{I}_w$, $w = 1, \dots, N_c$, be disjointed sets of indexes, at a given time t_0 , each one related to agents at collision risk, that is,

$$\|z_i - z_j\| \rightarrow 0 \ll \epsilon, \forall i, j \in \mathcal{I}_w, \forall w = 1, \dots, N_c.$$

The dynamics of two agents $i, j \in \mathcal{I}_w$ are then

$$\begin{aligned} \dot{z}_i(t) &= -z_i(t) - \frac{1}{n} \sum_{r=1}^n M \frac{z_i(t) - z_r(t)}{\|z_i(t) - z_r(t)\|^\nu + \epsilon} + u(t) \\ \dot{z}_j(t) &= -z_j(t) - \frac{1}{n} \sum_{s=1}^n M \frac{z_j(t) - z_s(t)}{\|z_j(t) - z_s(t)\|^\nu + \epsilon} + u(t) \end{aligned}$$

where $u(t) \triangleq r(t) + \dot{r}(t) - \zeta(t)$.

Denote with $\xi_{i,j}(t) \triangleq z_i(t) - z_j(t)$ the distance between the two agents, then

$$\begin{aligned} \dot{\xi}_{i,j}(t) &= -\xi_{i,j}(t) \\ &\quad - \frac{M}{n} \left(\sum_{r=1}^n \frac{\xi_{i,r}(t)}{\|\xi_{i,r}(t)\|^\nu + \epsilon} - \sum_{s=1}^n \frac{\xi_{j,s}(t)}{\|\xi_{j,s}(t)\|^\nu + \epsilon} \right). \end{aligned}$$

In first approximation, it is possible to consider the dynamics of $\xi_{i,j}$ depending only on the agents in $\mathcal{I}_w$ since the terms $\xi_{i,q} \approx \xi_{j,q}$, $\forall q \notin \mathcal{I}_w$. Moreover, the term $\|\xi_{i,j}\|^\nu \approx 0$, $\forall i, j \in \mathcal{I}_w$ which means that

$$\dot{\xi}_{i,j}(t) \approx - \left(I_k + \frac{|\mathcal{I}_w|}{n\epsilon} M \right) \xi_{i,j}(t)$$

where $|\mathcal{I}_w|$ is the cardinality of the set $\mathcal{I}_w$.

According to above equation, the term $\xi_{i,j}(t)$ is divergent if the matrix $(I_k + [|\mathcal{I}_w|/n\epsilon]M)$ is Hurwitz.

The analysis can be repeated for all sets of agents at collision risk.

Remark 3: Note that the above analysis is only an approximative study to provide some ideas on the clashes avoidance. Without fairly assumptions on collision avoidance, the proposed strategy can be intended only as a trajectory planner for the dynamic of each agent and the local agent control has to be designed to track the reference trajectory while avoiding collisions.

V. SOME EXTENSIONS TO THE KINEMATIC MODEL

In model (1), the interactions among agents are encoded by a time constant fully connected undirected graph ensuring that each member knows the exact position of all the others.

The previously described results have been obtained under the assumptions of agents with no size and able to interact with all other swarm members. In this section, the proposed swarm kinematic model will be analyzed in the cases of agents with a finite-size and assuming interactions only among near agents.

A. Limited Connectivity Among Agents

Assume that each agent is able to communicate only with agents in a neighborhood of radius δ from it and assume that once two agents meet, they exchange information about all their neighbors; in other words, modeling the agents as the nodes of an undirected graph, each agent is assumed to be able to interact with the agents in its same graph connected component. Let $\mathcal{G}_s(t)$, $s = 1, \dots, n_G$, be the connected components of the undirected graph formed by the swarm agents at time t . Focusing on the component $\mathcal{G}_r(t)$, each agent in this component is ruled by

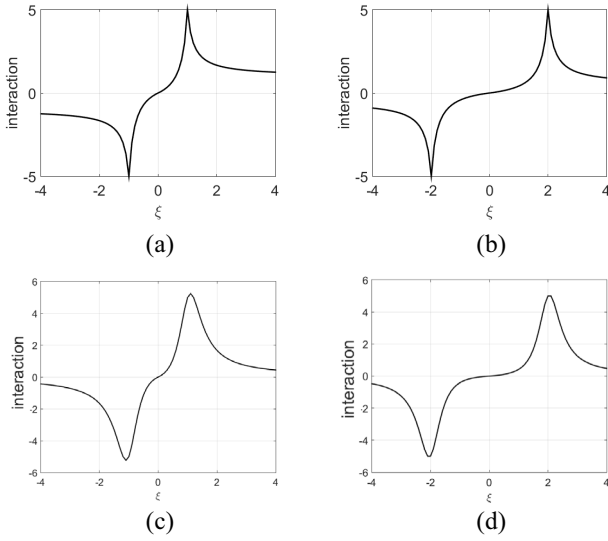
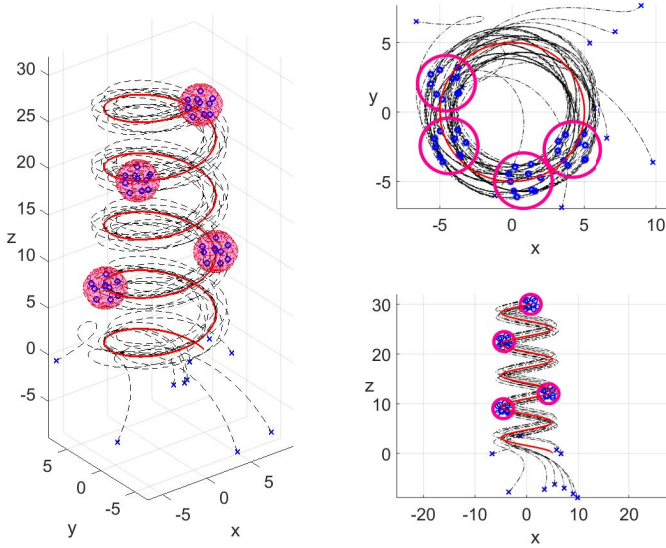
$$\begin{aligned} \dot{z}_i(t) &= -z_i(t) - \frac{1}{|\mathcal{G}_r(t)|} \sum_{j: z_j(t) \in \mathcal{G}_r(t)} \frac{M(z_i(t) - z_j(t))}{\|z_i(t) - z_j(t)\|^\nu + \epsilon} \\ &\quad + r(t) + \dot{r}(t) - \zeta_r(t) \end{aligned} \quad (8)$$

where $c_r(t)$ is the centroid of the agents which are in the connected component $\mathcal{G}_r(t)$, $|\mathcal{G}_r(t)|$ is the number of agents in this component, and $\zeta_r(t)$ is computed as $\zeta(t)$ but using $c_r(t)$ instead of $c(t)$. Each graph connected component is then seen as a stand-alone swarm of agents and once at least two agents of two different components meet, the two connected components become a single new one. Let τ be the time instant when two agents, one from $\mathcal{G}_r(\tau^-)$ and the other from $\mathcal{G}_s(\tau^-)$ meet, then $\mathcal{G}_r(\tau^-)$ and $\mathcal{G}_s(\tau^-)$ are connected creating the new component $\mathcal{G}_w(\tau)$. If $t < \tau$ the agents in the two components will be driven separately to $r(t)$ and to the hyper-ball of radius ρ around the reference trajectory. When $t \geq \tau$, the agents start to be driven by a new kinematic model, ruled by (8) related to the new connected component $\mathcal{G}_w(\tau)$, and the agents again will be driven to the same reference and to the same hyper-ball of the previous case, since $r(t)$ is independent from the connected component and ρ is independent from the number of agents into the swarm. It follows that once $\mathcal{G}_w(\tau)$ is formed, a new swarm of agents starts to evolve from the initial condition $z_i(\tau^-)$, $\forall z_i \in \mathcal{G}_w(\tau)$, and then all the results described in the previous sections are still valid if a limited connectivity among agents is assumed.

In a decentralized formulation the only assumption to be added is that in each connected component there is at least an agent able to detect the real external agent, evolving like $r(t)$. This agent will send this information to the remaining ones in the same connected component and they will then be able to obtain the required displacement information with respect to $r(t)$.

B. Finite-Size Agents

Assuming that the i th agent is centered in $z_i(t)$ its size is bounded by a hyper-ball of radius l , collisions among swarm members is avoided only if the distance among each couple of agents is always ensured to be greater than $d = 2l + \Lambda$, where Λ is a safety parameter. The kinematic model is modified to accomplish with this new constraint by adapting the interaction among agents so that to have high repulsion among them as their distance tends to d . To this end, the kinematic model

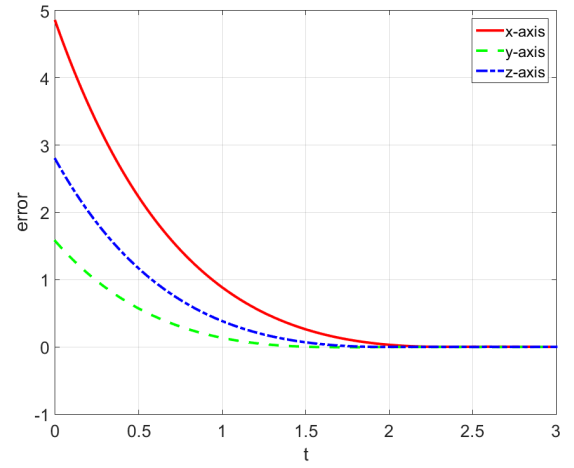
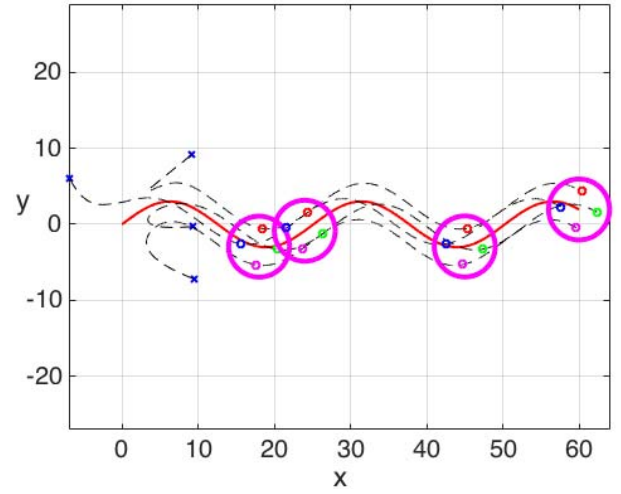
Fig. 3. Interaction effect with $\nu = 1, 2$, $d = 1, 2$, $\varepsilon = 0.01$.Fig. 4. Example 1: red line represents $r(t)$; crosses points are agents initial conditions; circles points are snapshots of agents positions at given time instants; and magenta spheres/circles represent instances of the hyper-ball where the agents converge in finite-time.

becomes

$$\dot{z}_i(t) = -z_i(t) - \frac{1}{n} \sum_{j=1}^n \frac{M(z_i(t) - z_j(t))}{d \left(\sqrt{(\|z_i(t) - z_j(t)\| - d)^2} \right)^\nu + \varepsilon} + r(t) + \dot{r}(t) - \zeta(t).$$

To highlight the interaction term properties, following the same lines described in Section II-B, consider a swarm formed by mono-dimensional agents, moving along the x -axis. The interaction term between two agents i and j , depending on $\xi(t) = x_i(t) - x_j(t)$, is

$$\frac{\xi(t)}{d \left(\sqrt{(|\xi(t)| - d)^2} \right)^\nu + \varepsilon}$$

Fig. 5. Example 1: error between $r(t)$ and $c(t)$.Fig. 6. Example 2: red line represents $r(t)$; crosses points are agents initial conditions; circles points are snapshots of agents positions at given time instants; and magenta circles represent instances of the hyper-ball where the agents converge in finite-time.

and it is shown in Fig. 3 for the cases $\nu = 1, 2$, $d = 1, 2$, and $\varepsilon = 0.2$. The interaction behavior in this latter case results in a shifted version of the standard interaction term described in Section II-B; when $|\xi(t)| \rightarrow d$ the interaction is $\mathcal{O}(1/\varepsilon)$. Obviously, in the case of limited connectivity, to ensure collision avoidance the detection field of each agent has to be greater than agents minimum allowed distance, i.e., $\delta \geq d$.

VI. NUMERICAL RESULTS

To test the proposed swarm kinematic model behavior three examples have been performed using the parameters $\nu = 1$, $\varepsilon = 10^{-6}$, and $\beta = 0.5$. In the presented simulation examples, a fully connected undirected graph has been used to define the swarm communication topology.

Example 1: A swarm formed by $n = 10$ agents in a 3-D space has been considered; $r(t)$ has been chosen as the spiral-like trajectory $r(t) = [5 \cos(t), 5 \sin(t), t]^T$ and $M = -2I_3$. The initial conditions have been randomly chosen in a sphere of radius 10 centered in the origin of $\mathbb{R}^3$. Figs. 4 and 5 show

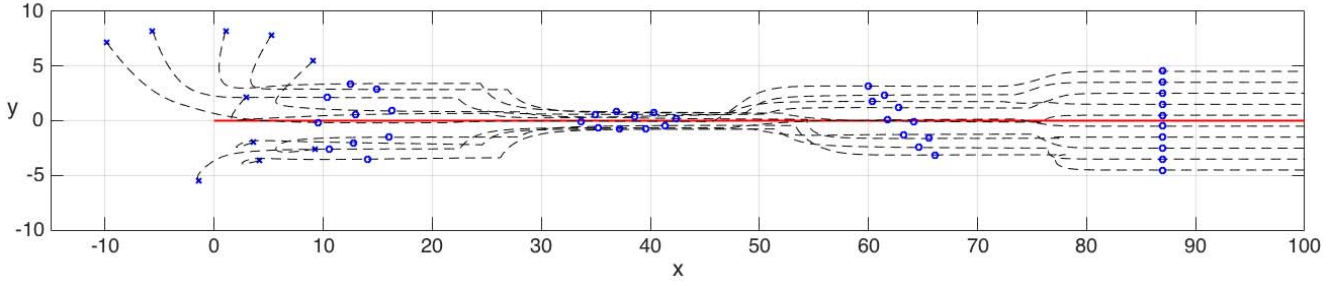


Fig. 7. Example 3: red line represents $r(t)$; crosses points are agents initial conditions; and circles points are snapshots of agents positions at given time instants.

the swarm evolution in terms of performed trajectory and the error $r(t) - c(t)$ between the desired trajectory and the swarm centroid. Note that the agents enter, and remain, into the defined hyper-ball (magenta ellipsoids and circles in Fig. 4 show some instances of this hyper-ball) around the centroid path and the centroid converges in finite-time to the reference trajectory.

Example 2: A swarm formed by four agents moving in a 2-D space has been used with $M = -4I_2$; $r(t)$ has been chosen as the sinusoidal-like signal $r(t) = [t, 3 \sin(t/4)]^T$. Fig. 6 shows the swarm performed trajectory highlighting that, as stated by Theorem 3, the agents reach a constant configuration with respect to the trajectory $r(t)$.

Example 3: This example aims at emphasizing the effects of the interaction matrix M on the swarm configuration. M has been chosen to modify the swarm shape during its evolution at every 25 s. A swarm of ten agents evolving in a 2-D space has been simulated. The agents start with a uniform repulsion effects by considering $M = -5I_2$; then the shape changes according to:

- 1) $M = -5 \begin{bmatrix} 1 & 0 \\ 0 & 0.5 \end{bmatrix}$ representing a higher repulsion on the first axis and resulting in an ellipsoid-like swarm shape having the greatest semiaxis along the first dimension;
- 2) $M = -5R \begin{bmatrix} 1 & 0 \\ 0 & 0.5 \end{bmatrix} R^T$ with R representing the rotation matrix of $\pi/4$ radians. This implies a related rotation of the previous ellipsoid shape;
- 3) $M = -5 \begin{bmatrix} 0.001 & 0 \\ 0 & 1 \end{bmatrix}$ which results in a straight line-like formation along the second axis.

The imposed reference trajectory is $r(t) = [t, 0]^T$ and the results are depicted in Fig. 7.

VII. CONCLUSION

This paper has presented a kinematic model for swarm formation with desired reference tracking capabilities. The dynamic of each agents is designed to guarantee that the agents aggregate in finite-time in a hyper-ball, the center of which tracks in finite-time a user-defined trajectory. The presence of an interaction matrix which mixes the agents coordinates allows to modify the form and the dimension of the aggregation region as highlighted in the numerical simulations. Under some hypotheses on this matrix, it has been proved that the

agents may also converge to a static configuration with respect to the swarm centroid and that they are ensured to avoid collisions with other swarm members.

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